

Advanced Machine Learning

Qing-Yuan Jiang

The School of Computer Science and Engineering, NJUST

2026.09.17

Basic Information

- ▶ Instructor: Qing-Yuan Jiang
- ▶ Research Interests: Machine Learning, Multimodal Learning.
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- ▶ Address: Room 4069, Dingxin Building.
- ▶ Homepage: <https://jiangqy.github.io/>

Course Information

- ▶ Course Page:
<https://jiangqy.github.io/course/>
- ▶ Location: Room III-514.
- ▶ TA: Peng Cheng
- ▶ Email: pengc@njust.edu.cn



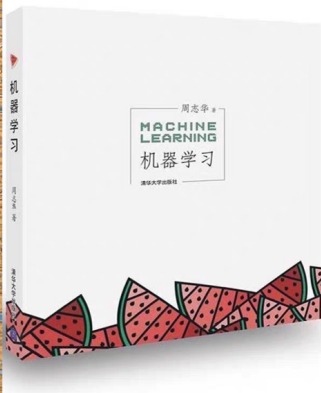
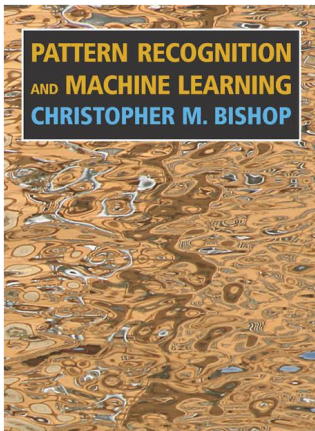
AML1班-2026秋季

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Text Books

- ▶ Pattern Recognition and Machine Learning, Bishop, C.M. & Nasrabadi, N.M.
- ▶ Machine Learning, Zhou ZH.



Course Assessment

Classroom Behavior: (20%)

- ▶ Attendance
- ▶ Question/Answer

Topic Report: (80%)

- ▶ Each student selects two candidate topics, one of which will be determined as final topic.
- ▶ Study the selected topic independently.
- ▶ Choose one presentation format:
 - ▶ English PPT Presentation: 20 minutes; students sign up
 - ▶ Chinese PPT Presentation: 20 minutes; students sign up
 - ▶ Video Presentation: 20 minutes; selected by the instructor and played in class

Topics

- ▶ Graph Neural Networks
- ▶ Variational Autoencoder
- ▶ Diffusion Model
- ▶ Self-Supervised Learning
- ▶ Online Learning
- ▶ Adversarial Learning
- ▶ Transfer Learning
- ▶ Contrastive Learning
- ▶ Retrieval-Augmented Generation

The presentation should introduce the selected topic from the perspectives of **its application scenarios, underlying principles, modeling, optimization.**

Important Dates

- ▶ Select a topic and decide on the presentation format:
2026.09.24
- ▶ Presentation: 2026.10.8-2026.10.30

THANKS